From Noise to Diversity: Random Embedding Injection in LLM Reasoning

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Abstract

Recent *soft prompt* methods, which prepend or insert trainable continuous embeddings into LLM inputs as virtual tokens, optimize these embeddings to enhance mathematical reasoning. However, the mechanism by which embedding injection itself affects model behavior, independent of optimization, remains unexplored. We investigate Random Soft Prompts (RSPs), continuous vectors sampled from an isotropic Gaussian fitted to the entrywise mean and variance of the pretrained embedding table and injected without any training. Theoretically, we show that the RSP contribution at each attention layer is exactly captured by a scalar random-token attention mass whose fixed-gap upper envelope shrinks as autoregressive cache growth accumulates real tokens, and that under a local affine analysis of the perturbed logits RSPs can assign positive probability to top-K entry regions that temperature sampling cannot create by rank-preserving scaling. Empirically, RSPs increase early-stage token entropy while generation stabilizes thereafter, and combining RSPs with temperature sampling yields higher trajectory diversity as measured by Pass@N. We further discuss the implications for GRPO training, where trajectory diversity is critical for learning signal quality.

1 Introduction

As large language models (LLMs) scale to billions of parameters, full fine-tuning becomes increasingly expensive. This has motivated parameter-efficient methods that adapt frozen models by introducing a small number of trainable parameters [Hu et al., 2022, Houlsby et al., 2019]. Among these, *soft prompts* prepend or insert learnable continuous embeddings into the input sequence, steering model behavior through the attention mechanism [Li and Liang, 2021, Lester et al., 2021]. This paradigm has been actively adopted for enhancing mathematical reasoning in LLMs [Kang et al., 2026, Xu et al., 2025, Ye et al., 2026, Hao et al., 2025, Zhang et al., 2026]. Despite their diverse designs, these methods share a common structure: they inject continuous embeddings that lie outside the pretrained vocabulary and optimize them to improve downstream performance.

However, we find that optimization is not a necessary condition for these effects. We observe that injecting randomly initialized, untrained continuous embeddings can also shift the output distribution. For instance, applying a chat template to a base model not fine-tuned with such format causes a prompt-format mismatch that degrades reasoning capability, but appending random embeddings to such mismatched inputs mitigates this degradation. If simple noise were merely perturbing the model, accuracy should worsen instead. This suggests that injecting out-of-vocabulary continuous embeddings itself, independent of any learned content, plays a non-trivial role in shaping model behavior. Existing studies primarily evaluate *soft prompt* methods through end-task accuracy, and the mechanisms by which embedding injection influences the output distribution remain insufficiently explored.

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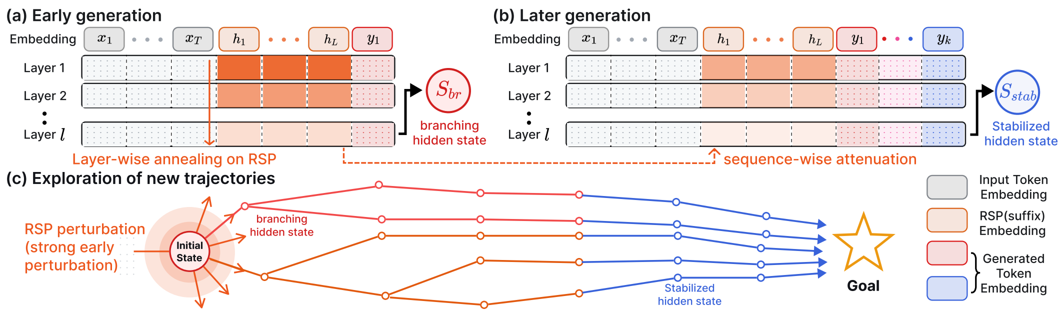

Figure 1: Conceptual overview of RSP-induced trajectory diversity (*suffix* injection). (a) Early decoding: RSP induces a branching hidden state (red) with a diverse output distribution. (b) Later decoding: cache growth attenuates the RSP attention mass, yielding a stabilized hidden state (blue). (c) As generation proceeds, the multiple trajectories opened by early RSP perturbation gradually stabilize into goal-directed completions.

In this work, we investigate the role of Random Soft Prompts (RSPs), continuous vectors sampled from an isotropic Gaussian fitted to the entrywise mean and variance of the pretrained embedding table and injected without any training. The role of randomness is not to imitate the embedding distribution; it is to ensure that repeated rollouts receive *independent* perturbations, which is what exploration requires. Theoretically, we identify an attention-level mechanism in which the RSP contribution is exactly captured by a scalar random-token attention mass whose fixed-gap upper envelope shrinks with autoregressive cache growth, and we show that under a local affine analysis of the perturbed logits RSPs can assign positive probability to top-K entry regions that temperature sampling cannot create by rank-preserving scaling. Empirically, RSPs increase early-stage token entropy while the output stabilizes in later generation stages, and combining RSPs with temperature sampling yields more diverse reasoning trajectories as measured by Pass@N scaling. We further discuss the implications of these findings for Group Relative Policy Optimization (GRPO) [Shao et al., 2024], where trajectory diversity is essential for effective learning signals.

Our contributions are as follows:

- We give a transformer-level mechanism in which the RSP contribution is controlled by a scalar attention mass with a cache-growth upper envelope, and a distributional analysis under a local affine surrogate showing that *independent isotropic resampling*—not a fixed perturbation—can reach top-K entry regions in a way temperature sampling cannot.
- We empirically analyze how RSPs affect LLM generation, showing that RSPs increase early token entropy and produce more diverse reasoning trajectories when combined with temperature sampling, and that the accumulation gain disappears when one RSP is shared across samples.
- We discuss the implications of RSP-induced diversity for GRPO training and argue that RSP, being rank-changing while temperature is rank-preserving, provides exploration gains complementary to existing temperature-based approaches.

2 Background

2.1 Soft Prompt Methods for LLM Reasoning

Soft prompts emerged as a parameter-efficient alternative to full fine-tuning. Prefix-Tuning [Li and Liang, 2021] introduced the paradigm by prepending trainable continuous vectors to every transformer layer, Prompt Tuning [Lester et al., 2021] simplified this to input-layer-only tuning and showed that it matches full fine-tuning at sufficient scale, and P-tuning [Liu et al., 2024] extended the approach with an LSTM-based prompt encoder to model inter-position dependencies.

More recently, *soft prompts* have been actively applied to LLM reasoning. TTSV [Kang et al., 2026] optimizes prefix embeddings at test time via entropy minimization, steering the model toward high-certainty states to activate latent reasoning capabilities. SoftCoT [Xu et al., 2025] replaces explicit chain-of-thought tokens with soft tokens generated by an assistant model, leveraging the stronger representational capacity of continuous representations over discrete language. LTPO [Ye et al., 2026] optimizes latent thought embeddings per test instance through policy gradient with a confidence-based intrinsic reward. COCONUT [Hao et al., 2025] feeds the model's own hidden states back as input embeddings, enabling reasoning in a continuous latent space. MemGen [Zhang et al., 2026] demonstrates that embedding vectors can serve as experience memory by constructing latent token sequences that enrich the model's reasoning. Pause tokens [Goyal et al., 2024] insert learnable tokens into the input to provide additional computation steps.

These approaches all inject out-of-vocabulary continuous embeddings optimized for task-specific objectives. Because evaluation centers on end-task accuracy, the contribution of the injection itself and that of the optimization are not separated, and the mechanisms by which such out-of-vocabulary continuous embeddings shape model behavior remain a separate and underexplored question.

2.2 Noise Injection in Neural Networks

Neural networks have incorporated noise at various stages. During training, dropout [Srivastava et al., 2014] randomly deactivates hidden units to prevent overfitting, and NEFTune [Jain et al., 2024] adds noise to token embeddings during instruction fine-tuning, reporting improvements in instruction following. These methods use noise as a regularizer and do not apply perturbations at inference time.

At inference time, perturbation-based approaches primarily target robustness and safety. Randomized smoothing [Cohen et al., 2019] injects Gaussian noise into inputs to obtain certified robustness guarantees. In the LLM setting, SmoothLLM [Robey et al., 2025] perturbs input prompts at the character level to defend against adversarial jailbreaking. These methods require multiple noisy forward passes with output aggregation, and target prediction stability rather than output diversity.

In reinforcement learning, NoisyNet [Fortunato et al., 2018] injects learnable noise into network weights to improve exploration, demonstrating that the location and structure of noise injection affect exploration quality. This motivation is closest to RSPs, which similarly target diversity, but differ in injection target (input embeddings vs. network weights) and require no learned noise parameters.

RSPs differ from these approaches in the following respects. **(1)** RSPs operate solely at inference without any training, whereas dropout and NEFTune inject noise during training, modifying the resulting model parameters accordingly. **(2)** RSPs preserve the original input and append fresh embedding vectors in a single forward pass, whereas robustness methods corrupt the input and aggregate over multiple noisy passes. **(3)** RSPs use purely random vectors sampled from the pretrained embedding statistics, whereas NoisyNet learns noise parameters through optimization.

3 Random Soft Prompts and Why They Work

This section states the mechanism used in the paper. A *rollout* means one full autoregressive decode under one sampled RSP. Within a rollout, an RSP enters the hidden state only through attention; the same scalar attention mass that enables an early influence is constrained by a cache-growth envelope as real tokens accumulate. Across rollouts, independent resampling can accumulate early branch differences into Pass@N gains under the task-side assumptions stated below. We keep the main statements and intuition here; further derivations are in Appendix D.

3.1 Random Soft Prompt

We start with the object being analyzed. Let $\mathbf{W}_E \in \mathbb{R}^{V\times d}$ denote the pretrained token-embedding matrix, where V is the vocabulary size and d is the hidden dimension. Write its entrywise mean and standard deviation as $\mu_E := \frac{1}{Vd}\sum_{i,k}[\mathbf{W}_E]_{i,k}$ and $\sigma_E := \mathrm{std}(\mathbf{W}_E)$. An RSP of length L is a sequence of continuous vectors $\mathbf{H} = [h_1, \ldots, h_L] \in \mathbb{R}^{L\times d}$ drawn independently from

$$h_j \sim \mathcal{N}(\mu_E \mathbf{1}_d,\ \sigma_E^2 \mathbf{I}_d), \quad j = 1, \ldots, L. \tag{1}$$

The centered form $\bar{h}_j := h_j - \mu_E \mathbf{1}_d \sim \mathcal{N}(0, \sigma_E^2 \mathbf{I}_d)$ is a zero-mean isotropic Gaussian. The RSP can be concatenated with the input embeddings $\mathbf{X}$ at one of three positions: *prefix* ($[\mathbf{H};\mathbf{X}]$), *infix* ($\mathbf{H}$ inserted within $\mathbf{X}$), or *suffix* ($[\mathbf{X};\mathbf{H}]$). These positions reflect the injection strategies in prior work: TTSV [Kang et al., 2026] prepends as *prefix*, SoftCoT [Xu et al., 2025] and LTPO [Ye et al., 2026] insert as *infix* between user and assistant turns, and MemGen [Zhang et al., 2026] appends as *suffix*. For repeated-sampling protocols, RSP draws and decoding randomness are independent across trials. The role of this randomness is not to imitate the embedding distribution but to give repeated rollouts *independent* perturbations, which is what exploration requires.

3.2 Exploration: how RSP enters the hidden state

We first ask how a single RSP injection perturbs the hidden state. RSP does not add to the hidden state directly; it enters only through attention weights, so its instantaneous influence within one head can collect into a single quantity: the attention mass assigned to RSP tokens.

Concretely, consider one attention head in self-attention layer ℓ with $n \geq 1$ unmasked real tokens, $L \geq 1$ RSP tokens, finite attention logits, attention weights $\alpha_{ij}^{(\ell)}$, and value vectors $\mathbf{v}_j^{(\ell)}, \mathbf{v}_{r_j}^{(\ell)}$ on the real and RSP sides. Define the RSP attention mass $w_{r,i}^{(\ell)} := \sum_{j=1}^{L} \alpha_{i,n+j}^{(\ell)}$. Since the real-token mass is then positive, the head output decomposes *exactly* as

$$\mathbf{o}_i^{(\ell)} \;=\; (1 - w_{r,i}^{(\ell)})\,\tilde{\mathbf{o}}_i^{(\ell)} \;+\; \boldsymbol{\eta}_i^{(\ell)}, \quad \tilde{\mathbf{o}}_i^{(\ell)} := \sum_{j=1}^{n} \frac{\alpha_{ij}^{(\ell)}}{1-w_{r,i}^{(\ell)}}\,\mathbf{v}_j^{(\ell)}, \quad \boldsymbol{\eta}_i^{(\ell)} := \sum_{j=1}^{L} \alpha_{i,n+j}^{(\ell)} \mathbf{v}_{r_j}^{(\ell)}. \tag{2}$$

Derivation of Eq. (2). Write the head output as $\sum_{j=1}^{n} \alpha_{ij}^{(\ell)} \mathbf{v}_j^{(\ell)} + \sum_{j=1}^{L} \alpha_{i,n+j}^{(\ell)} \mathbf{v}_{r_j}^{(\ell)}$, then factor $1 - w_{r,i}^{(\ell)} = \sum_{j=1}^{n} \alpha_{ij}^{(\ell)} > 0$ out of the real-token sum. No approximation beyond the softmax-normalization identity is used. □

The same scalar $w_{r,i}^{(\ell)}$ controls two things at once. The attenuation ratio $1 - w_{r,i}^{(\ell)}$ on the real-token signal and the upper bound $\|\boldsymbol{\eta}_i^{(\ell)}\| \leq w_{r,i}^{(\ell)} \, \max_j \|\mathbf{v}_{r_j}^{(\ell)}\|$ on the random contribution are both tied to it. The RSP-induced variation reduces to tracking a single number in $[0, 1]$. Early in decoding the cache is short and $w_{r,i}^{(\ell)}$ can be nontrivial, so the same identity that bounds the random contribution also lets one sampled RSP perturb early next-token decisions.

3.3 Annealing: KV cache growth dilutes RSP attention

The natural next question is how this scalar behaves as decoding proceeds. In autoregressive generation, the KV cache adds one real-token term at each step while the number of RSP terms stays fixed. Under a fixed real-vs-random logit gap, this cache growth gives an upper envelope.

Theorem 1 (Attention-mass decay under cache growth)**.** *For a query attending $n \geq 1$ real tokens and $L \geq 1$ RSP tokens with finite logits, let $\Delta_i^{(\ell)} := \min_{j\in[n]} \mathbf{q}_i^\top \mathbf{k}_j - \max_{j'\in[L]} \mathbf{q}_i^\top \mathbf{k}_{r_{j'}}$ denote the pre-scale logit gap between real and random tokens. In scaled dot-product attention,*

$$w_{r,i}^{(\ell)} \;\leq\; \frac{L}{L + n\,\exp(\Delta_i^{(\ell)}/\sqrt{d})},$$

and the right-hand side is strictly decreasing in n and tends to 0 at fixed L and $\Delta_i^{(\ell)}$.

Proof. Let $s_j := \mathbf{q}_i^\top \mathbf{k}_j/\sqrt{d}$, $s_{r_{j'}} := \mathbf{q}_i^\top \mathbf{k}_{r_{j'}}/\sqrt{d}$, $s_{r,\max} := \max_{j'} s_{r_{j'}}$, and $R := \sum_{j'=1}^{L} \exp(s_{r_{j'}})$. The gap definition gives $s_j \geq \Delta_i^{(\ell)}/\sqrt{d} + s_{r,\max}$ for every real j, and $\exp(s_{r,\max}) \geq R/L$ holds because the max dominates the average; combining the two and summing over the n real keys yields $\sum_{j=1}^{n} \exp(s_j) \geq (n/L)\exp(\Delta_i^{(\ell)}/\sqrt{d})\,R$. Substituting into $w_{r,i}^{(\ell)} = R/(\sum_j \exp(s_j) + R)$ gives

$$w_{r,i}^{(\ell)} \;\leq\; \frac{R}{(n/L)\exp(\Delta_i^{(\ell)}/\sqrt{d})\,R + R} \;=\; \frac{L}{L + n\,\exp(\Delta_i^{(\ell)}/\sqrt{d})}.$$

The derivative of the right-hand side in n is $-L\exp(\Delta_i^{(\ell)}/\sqrt{d})/(L + n\exp(\Delta_i^{(\ell)}/\sqrt{d}))^2 < 0$, so the bound is strictly decreasing and tends to 0 as $n \to \infty$. □

This single inequality carries two messages. The RSP length L is the design parameter we choose; it sits in the numerator and sets the *maximum* influence of the perturbation. The real-token count n grows automatically every step and sits in the denominator, so the *envelope* of the influence shrinks under the same L. What the theorem guarantees is the sequence-direction attenuation of the upper envelope at fixed (layer, query position, logit gap), not a layer-direction monotone decay; queries, keys, and hidden states co-evolve, so $\Delta_i^{(\ell)}$ also moves and the realized $w_{r,i}^{(\ell)}$ need not decrease at every step. The accurate reading is that "the envelope compatible with a fixed gap shrinks as n grows." Because Eq. (2) bounds the random contribution by the same scalar, the perturbation magnitude inherits the same envelope. This gives an implicit explore-then-exploit interpretation of decoding without asserting stepwise monotonic decay. Figure 2 verifies the sequence-direction prediction empirically.

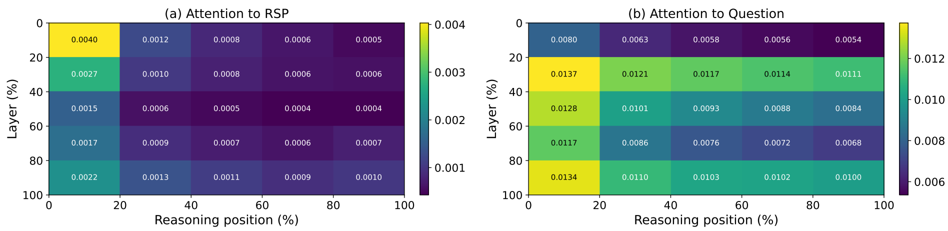

Figure 2: Per-token attention mass on Qwen2.5-Math-7B (suffix, 500 MATH-500 problems, each with an independently sampled RSP). (a) RSP-token attention decreases along the reasoning-position axis, consistent with the cache-growth bound of Theorem 1. The additional decrease along the layer axis is a separate empirical phenomenon suggesting that deeper layers sharpen the real-vs-random logit gap; it does not follow from Theorem 1 alone. (b) Question-token attention remains comparatively uniform. The reported quantity is the per-token mass of Appendix E, equal to $w_{r,i}^{(\ell)}/L$ averaged over heads and samples.

3.4 Performance gain: why independent reruns help

Theorem 1 explains what one rollout can do. The remaining question is performance: why can *fresh* RSP draws across reruns help more than repeating a single fixed perturbation? The answer needs one distributional fact about the prompt law and one separate task-side fact about productive branches.

The first fact is directional. The attention identity above does not require isotropy, but branch opening benefits from a perturbation law that does not under-serve whichever local direction matters for the current prompt. We state this fact for one centered RSP vector; the local surrogate below isolates one such perturbation from the length-L prompt. A minimax statement formalizes the point.

Theorem 2 (Direction-agnostic minimax coverage)**.** *Let D be any zero-mean distribution on $\mathbb{R}^d$ with covariance Σ_D and trace budget $\operatorname{tr}(\Sigma_D) \le \rho^2$. Then $\min_{\|b\|=1} \operatorname{Var}_{h\sim D}(b^\top h) = \lambda_{\min}(\Sigma_D) \le \rho^2/d$, with equality if and only if $\Sigma_D = (\rho^2/d)\mathbf{I}_d$.*

Equality holds for any distribution with isotropic covariance—Gaussianity is not the unique answer. The Gaussian RSP design *borrows the scale from the embedding distribution but discards its directional geometry on purpose*. The isotropic Gaussian additionally has positive density on all of $\mathbb{R}^d$ (full support), so it places no prior weight on any direction (Appendix C); learned fixed prompts ($\Sigma_D = 0$), PCA-aligned noise, vocabulary-token sampling, and embedding-covariance matching are all anisotropic and so under-explore at least one direction.

We now formalize branch opening on the surrogate. Linearizing the logits around $\bar{h} = 0$, define the affine surrogate $z_i^{\mathrm{lin}}(\bar{h}) := z_i + a_i^\top \bar{h}$ with $a_i := \nabla_{\bar{h}} z_i(0)$. For a baseline-excluded token i, the top-K entry region is $\mathcal{G}_{i,K} := \big\{\bar{h} \in \mathbb{R}^d : \#\{j \ne i : z_j^{\mathrm{lin}}(\bar{h}) > z_i^{\mathrm{lin}}(\bar{h})\} \le K-1\big\}$, matching the actual top-K definition.

Proposition 1 (Conditional top-K entry)**.** *If $\mathcal{G}_{i,K}$ contains a nonempty open set, the isotropic Gaussian's full support assigns it positive probability, so $\mu_i := \Pr_{\bar{h}}(\bar{h} \in \mathcal{G}_{i,K}) > 0$.*

This proposition gives a conditional reachability statement: once such a local top-K region exists in the surrogate, the RSP law assigns it positive probability. The existence of that region remains

Table 1: Accuracy (%) across model configurations and benchmarks. RSP reports the best result across injection positions (Prefix, Infix, Suffix). Full per-position breakdown is in Appendix A. Values in parentheses indicate the difference from the baseline.

	MATH-500		GSM8K		AIME24	
Model	Baseline	RSP	Baseline	RSP	Baseline	RSP
Instruct Models						
LLaMA-3.1-8B-Inst	52.20	49.40 (−2.8)	85.75	86.73 (+1.0)	6.67	10.00 (+3.3)
Qwen2.5-Math-7B-Inst	83.20	84.20 (+1.0)	95.45	95.68 (+0.2)	13.33	16.67 (+3.3)
Qwen2.5-Math-1.5B-Inst	73.00	75.20 (+2.2)	85.29	85.75 (+0.5)	10.00	20.00 (+10.0)
Base Models (Raw Text)						
Qwen2.5-Math-7B	72.20	71.60 (−0.6)	84.15	84.31 (+0.2)	6.67	16.67 (+10.0)
Qwen2.5-Math-1.5B	61.40	64.40 (+3.0)	77.86	74.30 (−3.6)	16.67	10.00 (−6.7)
Base Models + ChatML (Format Mismatch)						
Qwen2.5-Math-7B	52.20	70.40 (+18.2)	58.30	75.97 (+17.7)	23.33	23.33 (+0.0)
Qwen2.5-Math-1.5B	34.40	63.40 (+29.0)	37.45	67.02 (+29.6)	13.33	20.00 (+6.7)

a task-side condition, not a consequence of the mechanism alone. To lift a single-draw opening probability to task accuracy we add two task-side assumptions: (M1) each independent RSP run reaches a correct trajectory with marginal probability at least $p_{\min}(x) > 0$, and (M2) the N runs are mutually independent.

Proposition 2 (Multi-path Pass@N)**.** *Under (M1)–(M2), with N independent RSP runs,* $\Pr\big(\text{Pass@}N \textit{ on } x\big) \;\geq\; 1 - (1 - p_{\min}(x))^N \;\geq\; 1 - \exp(-N\, p_{\min}(x)).$

This division of labor is the main interpretive point. The mechanism can *admit* baseline-excluded candidates into the local candidate set when the corresponding entry region exists; whether an admitted branch is *productive* is task-dependent, which is why $p_{\min}(x) > 0$ is a separate assumption. Sharing one RSP across all samples removes this accumulation, so *independence* is essential. Section 4 tests exactly this signature.

4 Empirical Validation

We now examine how the explanation in ğ3 appears in actual LLM generation through three scenes. First, we ask whether RSP preserves baseline accuracy without breaking it, and recovers it in some settings (§4.2). Second, we check whether early diversification followed by late stabilization—the empirical signature suggested by the attention decomposition and cache-growth envelope—is observed (§4.3). Finally, we contrast Pass@N between sharing one RSP across samples versus drawing a fresh RSP per sample, directly probing whether *independence* is the key (§4.4).

4.1 Experimental Setup

We evaluate on three mathematical reasoning benchmarks, MATH-500 [Lightman et al., 2024], GSM8K [Cobbe et al., 2021], and AIME24, using LLaMA-3.1-8B-Instruct [Grattafiori et al., 2024] and the Qwen2.5-Math model family [Yang et al., 2024] across instruct models, base models, and base models with mismatched chat templates—seven configurations in total. All experiments use greedy decoding to isolate the effect of RSPs from sampling stochasticity, and the answer-extraction pipeline uses fallbacks to score only the final answer. RSPs follow the definition of ğ3.1 and are concatenated at one of three positions (prefix, infix, suffix) with a freshly drawn $\mathbf{H}$ for each batch. Full experimental details and per-position results are in Appendix A.

4.2 How Does Untrained RSP Compare to Baselines & Optimized Soft Prompts?

We first examine how injecting random embeddings affects the model's reasoning performance. Table 1 reports the accuracy for each model configuration, using the best RSP result across injection positions (Prefix, Infix, Suffix) and $L \in \{10, 15, 20\}$ for each benchmark.

Table 2: Comparison with existing soft prompt methods (TTSV, SoftCoT, LTPO) under unified evaluation. Baseline and RSP values are from Table 1. **Bold** denotes the best result and underline denotes the second best for each model–benchmark pair.

Model	Benchmark	Baseline	RSP	TTSV	SoftCoT	LTPO
Qwen2.5-Math-7B-Instruct	MATH-500	83.20	**84.20**	83.80	82.80	83.00
	GSM8K	95.45	**95.68**	95.30	95.00	94.84
	AIME24	13.33	16.67	**23.30**	16.67	13.33
Qwen2.5-Math-1.5B-Instruct	MATH-500	73.00	75.20	75.20	**76.00**	72.40
	GSM8K	85.29	**85.75**	85.70	84.46	84.53
	AIME24	10.00	**20.00**	6.70	6.67	10.00

For instruct models and base models with their native prompt formats, RSP injection maintains or slightly shifts baseline performance within a few percentage points. Most strikingly, under prompt-format mismatch — where base models receive ChatML templates they were not trained on — RSP yields pronounced improvements of up to +29 pp. These gains are difficult to attribute to simple perturbation effects, as noise that merely raises model uncertainty would further degrade performance.

We compare RSPs with prior soft prompt methods (TTSV, SoftCoT, LTPO) that optimize their embeddings for distinct objectives, under a unified evaluation protocol reported in Table 2. All methods are re-evaluated with the same unified extraction pipeline (Appendix A.2), enabling a format-agnostic comparison despite each method's distinct prompt format and grading scheme. The results place RSPs on par with the optimized methods without requiring any training.

4.3 How Does RSP Affect the Output Distribution?

We analyze how RSP injection changes the model's output distribution on Qwen2.5-Math-1.5B-Instruct with MATH-500, measuring per-token entropy, top-1 probability, and varentropy across the generation process. Figure 3 compares these metrics for the first 5% of generation steps across RSP variants and prior soft prompt methods (LTPO, SoftCoT, TTSV). Full statistics are in Appendix A.4.

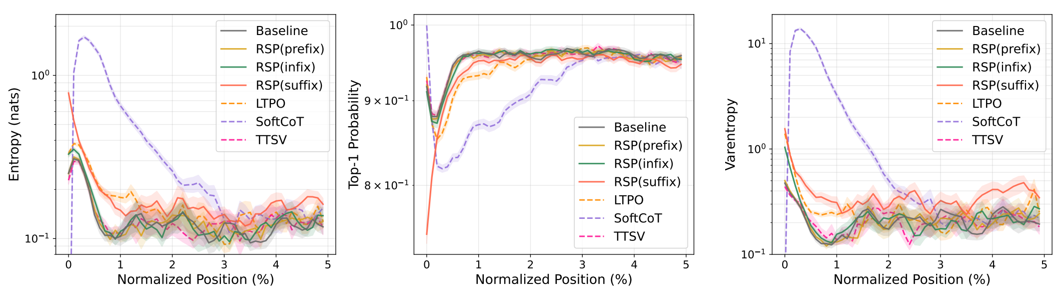

Figure 3: Mean entropy, top-1 probability, and varentropy during the first 5% of generation steps (Qwen2.5-Math-1.5B-Instruct, MATH-500). Shading indicates ±1 SEM. Solid lines: RSP variants and the baseline. Dashed lines: prior soft prompt methods (LTPO, SoftCoT, TTSV).

These results show that RSP modifies the early-stage output distribution across all injection positions: entropy increases (the token distribution flattens), top-1 probability decreases (commitment to a dominant token weakens), and varentropy increases (indicating multimodal distributions [Ahmed et al., 2026]). High-entropy positions are known to act as critical forking points that determine reasoning trajectories [Wang et al., 2025], so this shift moves the model toward a state where multiple reasoning paths can coexist. The change is localized to the initial reasoning stage. In the middle and final portions of generation, all three metrics converge to baseline levels.

Although existing soft prompt methods are optimized for different objectives, they share a common pattern of increasing early-stage entropy. Notably, TTSV, which optimizes soft prompts using a reward that reduces the mean entropy of reasoning trajectories, still exhibits early entropy comparable to the baseline. This suggests that elevating early-stage entropy is an inherent effect of injecting non-pretrained continuous embeddings, independent of the optimization objective.

4.4 Does RSP Induce Trajectory Diversity?

Section 4.3 empirically established that RSP reshapes the early-stage output distribution. We now examine whether this shift translates into reasoning diversity at three complementary levels of analysis: token rankings, semantic content of trajectories, and task-level outcomes. The three analyses converge on a directional picture: RSP's first-token perturbation propagates into semantically more diverse reasoning trajectories, which in turn yields greater output diversity at the task level. All three analyses share the setting: Qwen2.5-Math-1.5B-Instruct, MATH-500, suffix injection, and $L = 10$.

Token-level: distribution beyond temperature. We quantify how much RSP shifts the first-token distribution, using temperature sampling as a baseline. Under autoregressive decoding, any trajectory-level divergence between RSP and the baseline must originate from the first generation step, a critical forking point for reasoning trajectories [Wang et al., 2025]. With the baseline at τ=1.0 as reference, we compare baselines at τ=2.0 and τ=3.0 (16 samples per problem) against RSP at τ=1.0 (averaged over 16 seeds) on four metrics: Spearman rank correlation ρ measuring whether the relative ranking of tokens is preserved, probability mass outside the baseline's top-10 measuring support expansion to low-probability tokens, Jensen–Shannon divergence as the overall distributional distance, and Pass@1 accuracy (formal definitions in Appendix B).

Table 3: First-token distribution metrics measured against the baseline at τ=1.0, comparing baselines at higher temperatures (τ=2.0, 3.0) with RSP at τ=1.0. RSP metrics are averaged over 16 random seeds; Acc reports mean $\pm$ std across 16 samples per problem.

Metric	τ=2.0	τ=3.0	RSP
Spearman ρ	1.000	1.000	**0.709**
Mass outside top-10	5.18%	29.11%	**21.86%**
JS divergence	0.060	0.218	**0.131**
Acc (%)	20.77 $\pm$ 1.47	1.42 $\pm$ 0.35	**73.30 $\pm$ 1.07**

Table 3 reports the contrast. RSP's distributional shift falls between τ=2.0 and τ=3.0 in magnitude (mass outside top-10 and JS divergence), yet its mechanism and cost differ sharply. Temperature scaling is a monotone transform that preserves token ranking, so ρ relative to the baseline is exactly 1 at any τ. RSP partially preserves the ranking but induces reranking, yielding $\rho = 0.709$. The Pass@1 row exposes the cost: approaching RSP's magnitude requires raising τ toward 3.0, but accuracy collapses to 1.42%, while RSP preserves accuracy at 73.30%. Rather than uniformly flattening the distribution as temperature scaling does, RSP produces a fundamentally different kind of perturbation.

Trajectory-level: semantic diversity. Token-level reranking raises a second question: do first-token perturbations produce semantically diverse reasoning trajectories, or do independently perturbed trajectories converge to semantically similar ones? We randomly sample 64 problems from MATH-500 and generate 300 independent trajectories per problem under Baseline and RSP at temperature $\tau = 1.0$. Each trajectory is encoded into a dense semantic vector by BGE-M3 [Chen et al., 2024], a sentence-level multilingual embedding model, and we compute three complementary metrics over these embeddings: *pairwise cosine distance* between trajectories captures the overall semantic spread of the trajectory set; *inter-cluster distance* between centroids identified by DBSCAN [Ester et al., 1996], a density-based clustering algorithm that groups nearby trajectories without prespecifying the number of clusters, captures the separation between distinct trajectory groups; and *intra-cluster distance* captures semantic coherence within each group.

Table 4: Semantic diversity metrics (64 problems, 300 trajectories per condition).

Metric	Baseline	RSP
Pairwise cos. dist.	0.0612	**0.0879**
Inter-cluster dist.	0.0183	**0.0982**
Intra-cluster dist.	0.0526	0.0528

Table 4 reveals three concurrent patterns. Pairwise cosine distance rises by $1.4\times$, indicating that the trajectory set as a whole becomes more semantically diverse. Inter-cluster distance jumps by $5.4\times$, showing that the trajectories split into substantially more separated groups. Intra-cluster distance is essentially unchanged, indicating that semantic coherence within each group remains comparable to the baseline. RSP also yields larger pairwise distance than baseline on 56 of 64 problems (87.5%).

Outcome-level: Pass@N scaling. Finally, we evaluate whether the token- and trajectory-level diversity translates into task-level outcome diversity through Pass@N [Chen et al., 2021], the probability that at least one of N sampled solutions is correct. We compare four conditions with $N = 16$ samples per problem on MATH-500 and AIME24: *(i) Baseline,* $\tau = 1.0$: temperature sampling only; *(ii) RSP (fixed seed),* $\tau = 1.0$: a single RSP shared across all samples; *(iii) RSP (indep.*

seed), greedy: a different RSP per sample without temperature; and *(iv) RSP (indep. seed),* $\tau = 1.0$: a different RSP per sample combined with temperature sampling.

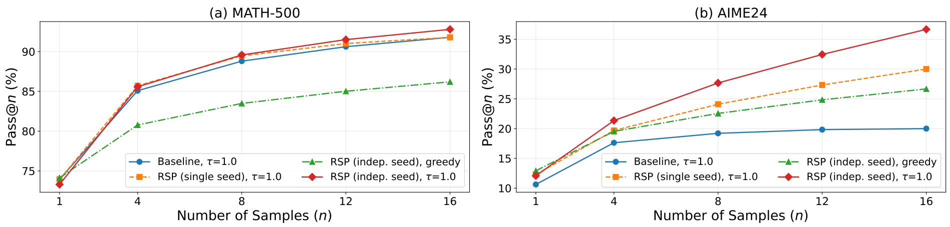

Figure 4: Pass@N scaling on (a) MATH-500 and (b) AIME24 with Qwen2.5-Math-1.5B-Instruct, $N = 16$ samples per problem. Baseline: temperature sampling only. Fixed seed: single RSP shared across samples combined with temperature. Indep. seed: a different RSP per sample, with or without temperature.

Condition *(iv)* achieves the highest Pass@N on both benchmarks, reaching 92.80% on MATH-500 and 36.67% on AIME24 at $N = 16$. Condition *(ii)* shows no improvement over the baseline, suggesting that the diversity gain likely depends on varying the random embeddings across samples rather than a single fixed perturbation. Pass@1 remains comparable to the baseline on MATH-500, and on the more challenging AIME24, *(iv)* achieves higher Pass@1 than the baseline. Together these observations suggest that combining independent RSPs with temperature sampling may yield greater trajectory diversity while preserving single-sample accuracy.

5 Application: DAPO Training with RSP

Sections 3–4 establish what RSP does (attention-mediated branch opening) and how it behaves at inference (Pass@N gains, rank-changing diversity unattainable by temperature). The remaining question is whether these effects transfer to training-time performance gains. Rollout diversity is a central determinant of learning in sampling-based RL, and DAPO [Yu et al., 2025] is a recent GRPO [Shao et al., 2024] variant explicitly designed to preserve that diversity at the loss level. We use it as a stringent testbed: does input-level branch opening compose with loss-level diversity preservation to yield further gains? Implementation details and full per-step results are deferred to Appendix F.

DAPO extends GRPO with loss-term modifications that preserve rollout diversity during long chain-of-thought RL training, shaping diversity at the loss level by reshaping how rollouts are weighted and clipped. RSP, in contrast, induces diversity at the input level by perturbing the rollout distribution before any loss is computed (Section 4.4). Let o_i be the i-th rollout response, $r_{i,t}(\theta) = \pi_\theta(o_{i,t} \mid q, [\mathbf{H}_g,] o_{i,<t}) / \pi_{\theta_{\text{old}}}(o_{i,t} \mid q, [\mathbf{H}_g,] o_{i,<t})$ the importance ratio, $\hat{A}_{i,t}$ the group-relative advantage, and $(\varepsilon_{\text{low}}, \varepsilon_{\text{high}})$ the asymmetric clip range, with the bracketed $\mathbf{H}_g$ denoting the RSP appended to the prompt. The DAPO + RSP objective is

$$\mathcal{J}_{\text{DAPO+RSP}}(\theta) = \mathbb{E}\left[\frac{1}{\sum_i |o_i|} \sum_{i,t} \min\Big(r_{i,t}\hat{A}_{i,t},\, \text{clip}\big(r_{i,t},\, 1-\varepsilon_{\text{low}},\, 1+\varepsilon_{\text{high}}\big)\hat{A}_{i,t}\Big)\right]. \quad (3)$$

Table 5: Per-benchmark accuracy (%) at each method's peak step on Qwen2.5-Math-7B. **Bold** denotes the better of the two RL methods on each benchmark. Avg is the unweighted five-benchmark mean.

Method	GSM8K	MATH-500	College	Minerva	AIME 24	Avg
Base	57.2	52.6	18.3	13.6	8.6	30.06
DAPO	86.3	78.2	41.4	37.5	22.6	53.20
DAPO + RSP	**87.7**	**78.6**	**42.2**	**39.7**	**23.6**	**54.36**

We train Qwen2.5-Math-7B on a MATH Level-3–5 subset (~8.5K prompts) using the SimpleRL-Zoo [Zeng et al., 2025] training pipeline with the objective of Eq. (3). Each step samples $G = 8$ rollouts per prompt on a batch of 1,024 prompts under τ=1.0, followed by 4 PPO mini-batch updates of 256 prompts each, for 100 training steps (about 12 epochs). We compare *DAPO* against *DAPO + RSP*, where RSP *suffix* injection ($L = 20$ tokens) is applied during rollouts only; evaluation is performed without RSP. We evaluate on five mathematical-reasoning benchmarks (GSM8K [Cobbe et al., 2021], MATH-500 [Lightman et al., 2024], College Math, Minerva Math [Lewkowycz et al., 2022], AIME 2024). AIME 2024 is scored by Avg@32 at τ=1.0 to reduce variance, while the larger benchmarks use greedy decoding. The reported average is the unweighted mean over these five benchmarks.

A first observation in Figure 5 and Table 5 is the consistent advantage of DAPO + RSP at the upper end of training: the five-benchmark average accuracy rises to 54.36% at step 90, exceeding DAPO's own peak of 53.20% at step 80 by +1.16 pp; the gap widens at step 100 to 53.64% versus 50.54% (+3.10 pp), and the post-step-80 decline observed under DAPO alone is delayed when RSP is combined. This pattern follows from the fact that RSP and DAPO's loss-term modifications act at separated stages of the optimization. DAPO's modifications take already-generated rollouts and logits as input and preserve diversity at the loss level, whereas RSP modifies the hidden states from which those rollouts are produced (Section 3.2). The additional gain observed on top of DAPO's aggressive loss-level diversity preservation suggests that hidden-state-level perturbation operates on a surface that loss-level preservation does not reach.

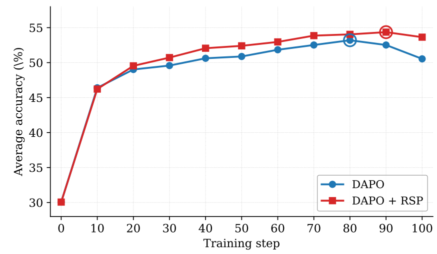

Figure 5: Five-benchmark average accuracy on Qwen2.5-Math-7B. DAPO + RSP reaches a higher peak (step 90) and stays stable through step 100.

A second observation is that the two curves cross around step 20, with DAPO + RSP showing slower reward growth in the earlier phase. This crossing is consistent with the multi-path argument of Section 3.4 reproduced in training dynamics, where deviations from the trajectories the policy would otherwise have selected expose the learning signal to a broader trajectory distribution. The cumulative effect of that broader exposure manifests as the later-stage ceiling improvement and delayed collapse. The same pattern has long been documented as the exploration–exploitation dynamics of reinforcement learning [Sutton and Barto, 2018]; the present result shows that this dynamic can also be induced from the input side through hidden-state-level perturbation.

6 Conclusion

We studied Random Soft Prompts (RSPs), isotropic Gaussian vectors fitted to the entrywise mean and variance of the pretrained embedding table and injected without training. Theoretically, the RSP contribution at each attention layer is exactly captured by a scalar attention mass whose fixed-gap upper envelope shrinks under cache growth, and under a local affine analysis RSPs can reach top-K entry regions that rank-preserving temperature scaling cannot create (§3). Empirically, RSPs raise early-stage entropy while later generation stabilizes, and the accumulation gain disappears when one RSP is shared across samples (§4). Because RSP is rank-changing while temperature is rank-preserving, the two are complementary, suggesting a sampling-level remedy for entropy collapse in GRPO [Shao et al., 2024] beyond DAPO [Yu et al., 2025] and CE-GPPO [Su et al., 2026]; training-time validation, extension beyond mathematical reasoning, and a principled choice of RSP length and injection position—together with the task-side bound $p_{\min}(x)>0$ of Proposition 2—are natural next steps.

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A Experimental Setup and Full Accuracy Breakdown

Table 6 provides the full per-position accuracy breakdown (Prefix, Infix, Suffix) corresponding to the best-across-positions results summarized in Table 1 in the main text. Table 7 lists the number of RSP tokens L used for each model–benchmark pair, selected from $\{10, 15, 20\}$. All experiments are conducted on a single NVIDIA A6000 40GB GPU with greedy decoding, a maximum generation length of 3,072 tokens for MATH-500 and GSM8K, and 4,096 tokens for AIME24. Batch size is fixed per model (16 for LLaMA-3.1-8B-Instruct and Qwen2.5-Math-7B variants, 32 for Qwen2.5-Math-1.5B variants).

Table 6: Accuracy (%) across model configurations, injection positions, and benchmarks. Values in parentheses indicate the difference from the baseline. **Bold** denotes the best result and <u>underline</u> denotes the second best for each model–benchmark pair.

Model	Mode	MATH-500	GSM8K	AIME24
Instruct Models				
LLaMA-3.1-8B-Instruct	Baseline	**52.20**	85.75	<u>6.67</u>
	Prefix	45.00 (−7.2)	76.35 (−9.4)	3.33 (−3.3)
	Infix	<u>49.40</u> (−2.8)	<u>85.97</u> (+0.2)	**10.00** (+3.3)
	Suffix	<u>49.40</u> (−2.8)	**86.73** (+1.0)	**10.00** (+3.3)
Qwen2.5-Math-7B-Instruct	Baseline	83.20	95.45	<u>13.33</u>
	Prefix	81.40 (−1.8)	94.92 (−0.5)	<u>13.33</u> (+0.0)
	Infix	**84.20** (+1.0)	<u>95.60</u> (+0.2)	**16.67** (+3.3)
	Suffix	<u>83.40</u> (+0.2)	**95.68** (+0.2)	**16.67** (+3.3)
Qwen2.5-Math-1.5B-Instruct	Baseline	73.00	<u>85.29</u>	10.00
	Prefix	<u>74.80</u> (+1.8)	<u>85.29</u> (+0.0)	<u>13.33</u> (+3.3)
	Infix	**75.20** (+2.2)	**85.75** (+0.5)	6.67 (−3.3)
	Suffix	74.20 (+1.2)	84.69 (−0.6)	**20.00** (+10.0)
Base Models (Raw Text)				
Qwen2.5-Math-7B	Baseline	**72.20**	<u>84.15</u>	6.67
	Prefix	<u>71.60</u> (−0.6)	**84.31** (+0.2)	**16.67** (+10.0)
	Infix	63.20 (−9.0)	64.29 (−19.9)	**16.67** (+10.0)
	Suffix	55.60 (−16.6)	54.13 (−30.0)	<u>13.33</u> (+6.7)
Qwen2.5-Math-1.5B	Baseline	<u>61.40</u>	**77.86**	**16.67**
	Prefix	**64.40** (+3.0)	<u>74.30</u> (−3.6)	<u>10.00</u> (−6.7)
	Infix	63.20 (+1.8)	73.92 (−3.9)	<u>10.00</u> (−6.7)
	Suffix	54.60 (−6.8)	58.61 (−19.3)	3.33 (−13.3)
Base Models + ChatML (Format Mismatch)				
Qwen2.5-Math-7B	Baseline	52.20	58.30	**23.33**
	Prefix	59.20 (+7.0)	56.41 (−1.9)	<u>3.33</u> (−20.0)
	Infix	<u>62.00</u> (+9.8)	<u>69.07</u> (+10.8)	**23.33** (+0.0)
	Suffix	**70.40** (+18.2)	**75.97** (+17.7)	**23.33** (+0.0)
Qwen2.5-Math-1.5B	Baseline	34.40	37.45	<u>13.33</u>
	Prefix	**63.40** (+29.0)	**67.02** (+29.6)	**20.00** (+6.7)
	Infix	39.20 (+4.8)	50.42 (+13.0)	6.67 (−6.7)
	Suffix	<u>51.00</u> (+16.6)	<u>50.64</u> (+13.2)	<u>13.33</u> (+0.0)

Table 7: Number of RSP tokens (L) used for each model and benchmark.

Model	MATH-500	GSM8K	AIME24
LLaMA-3.1-8B-Instruct	15	20	15
Qwen2.5-Math-7B-Instruct	20	20	20
Qwen2.5-Math-1.5B-Instruct	20	10	20
Qwen2.5-Math-7B	10	10	20
Qwen2.5-Math-1.5B	10	10	20
Qwen2.5-Math-1.5B (ChatML)	20	20	10
Qwen2.5-Math-7B (ChatML)	20	20	20

A.1 RSP Injection Positions and Prompt Templates

Below we show the prompt structure for each injection position across all model types. Blue text indicates RSP embeddings, and purple text indicates special tokens.

A.1.1 Prefix

RSP embeddings are concatenated before the prompt embeddings. No text modification is applied.

ChatML (Qwen Instruct / Base + ChatML).

```
[Random Embeddings (L tokens)]
<|im_start|>system
Please reason step by step, and put your final answer within \boxed{}.<|im_end|>
<|im_start|>user
{question}<|im_end|>
<|im_start|>assistant
```

LLaMA Chat Template.

```
[Random Embeddings (L tokens)]
<|begin_of_text|>
<|start_header_id|>system<|end_header_id|>
Cutting Knowledge Date: December 2023
Today Date: 26 Jul 2024
Please reason step by step, and put your final answer within \boxed{}.<|eot_id|>
<|start_header_id|>user<|end_header_id|>
{question}<|eot_id|>
<|start_header_id|>assistant<|end_header_id|>
```

Raw Text (Qwen Base).

```
[Random Embeddings (L tokens)]
{question}
Please reason step by step, and put your final answer within \boxed{}.
```

A.1.2 Infix

A description text and special tokens are inserted into the prompt. The special token positions are then replaced with RSP embeddings at the embedding level.

ChatML (Qwen Instruct / Base + ChatML).

```
<|im_start|>system
Please reason step by step, and put your final answer within \boxed{}.<|im_end|>
<|im_start|>user
{question}
There are {L} special tokens that contain compressed latent reasoning information that
might be useful for your reasoning. If these tokens are useful for your case, you can
use them as reference. If these tokens are not useful for your case, you can ignore
them and focus back to solving the problem.
Here are the {L} special tokens: <special_token_1>...<special_token_L>
<|im_end|>
<|im_start|>assistant
```

LLaMA Chat Template.

```
<|begin_of_text|>
<|start_header_id|>system<|end_header_id|>
Cutting Knowledge Date: December 2023
Today Date: 26 Jul 2024
Please reason step by step, and put your final answer within \boxed{}.<|eot_id|>
<|start_header_id|>user<|end_header_id|>
```

```
{question}
There are {L} special tokens that contain compressed latent reasoning information that
might be useful for your reasoning. If these tokens are useful for your case, you can
use them as reference. If these tokens are not useful for your case, you can ignore
them and focus back to solving the problem.
Here are the {L} special tokens:
<reserved_special_token_0>...
<reserved_special_token_L>
<|eot_id|>
<|start_header_id|>assistant<|end_header_id|>
```

Raw Text (Qwen Base).

```
{question}
There are {L} special tokens that contain compressed latent reasoning information that
might be useful for your reasoning. If these tokens are useful for your case, you can
use them as reference. If these tokens are not useful for your case, you can ignore
them and focus back to solving the problem.
Here are the {L} special tokens: <special_token_1>...<special_token_L>
Please reason step by step, and put your final answer within \boxed{}.
```

A.1.3 Suffix

For chat-templated models, RSP embeddings are inserted immediately before the end-of-turn token. For raw text models, RSP embeddings are concatenated at the end of the prompt.

ChatML (Qwen Instruct / Base + ChatML).

```
<|im_start|>system
Please reason step by step, and put your final answer within \boxed{}.<|im_end|>
<|im_start|>user
{question}[Random Embeddings (L tokens)]<|im_end|>
<|im_start|>assistant
```

LLaMA Chat Template.

```
<|begin_of_text|>
<|start_header_id|>system<|end_header_id|>
Cutting Knowledge Date: December 2023
Today Date: 26 Jul 2024
Please reason step by step, and put your final answer within \boxed{}.<|eot_id|>
<|start_header_id|>user<|end_header_id|>
{question}[Random Embeddings (L tokens)]<|eot_id|>
<|start_header_id|>assistant<|end_header_id|>
```

Raw Text (Qwen Base).

```
{question}
Please reason step by step, and put your final answer within \boxed{}.[Random
Embeddings (L tokens)]
```

A.2 Answer Extraction and Grading

The final answer is extracted from the model output through a fallback chain. Each step is attempted in order; if extraction fails, the next step is tried.

Answer Extraction Fallback Chain

1. `"final answer is $...$. I hope"` pattern (Minerva format)
2. `\boxed{...}`: last non-empty box is used; empty `\boxed{}` are skipped
3. Text following `"the answer is"`
4. Text following `"final answer is"`
5. Last number in the output (regex fallback)

Extracted answers are normalized by removing `$`, `\left`, `\right`, and unit strings. Grading is performed via symbolic equivalence checking: exact string match, numerical comparison (`math.isclose`, `rel_tol=1e-4`), and SymPy symbolic simplification.

A.3 Reproduced Prior Work Hyperparameters

All prior methods are reproduced on Qwen2.5-Math-1.5B-Instruct and Qwen2.5-Math-7B-Instruct, evaluated with greedy decoding (temperature $= 0$) and our unified answer extraction pipeline.

TTSV. Prefix length 20, trained for 20 epochs with AdamW optimizer, batch size 2, gradient accumulation 8 steps, and linear learning rate schedule. Learning rate is 1e-3 for Qwen-1.5B and 1e-5 for Qwen-7B.

LTPO. Table 8 reports the per-benchmark hyperparameters.

Table 8: LTPO hyperparameters.

Parameter	GSM8K	MATH-500	AIME24
tokens	8	8	8
steps	20	20	20
top_k	10	10	10
sigma	20	20	5
sigma_decay	0.95	0.95	0.95
lr	1e-2	5e-3	5e-2

SoftCoT. Projection module trained for 10 epochs. Thought tokens: 32 during training, 4 during evaluation. Batch size 4 for GSM8K, 1 for MATH with gradient accumulation 4. The small (assistant) model is Qwen2.5-Math-1.5B-Instruct in both configurations, while the large model is Qwen2.5-Math-1.5B-Instruct (learning rate 2e-5) or Qwen2.5-Math-7B-Instruct (learning rate 5e-6). For MATH-500 and AIME24 evaluation, we use the projection module trained on GSM8K, as it yields higher accuracy than the one trained on MATH.

A.4 Full Entropy Curves

Figure 6 shows the full normalized (0–100%) generation trajectories for entropy, top-1 probability, and varentropy, and Table 9 reports the corresponding scalar statistics including overall means, first-5% means, and the average number of generated tokens per problem. All methods converge to similar levels after the initial generation stage, confirming that the distributional changes induced by RSP are localized to the early reasoning phase.

Table 9: Per-step statistics for Qwen2.5-Math-1.5B-Instruct on MATH-500 across the full generation and the first 5% of generation steps, together with the average number of generated tokens per problem. Top-1 Prob (overall) is reported as a weighted average over the first-10%/middle/last-10% segment means with weights $0.1/0.8/0.1$. Extension of Figure 3 (main text) and Figure 6.

Metric	Baseline	Prefix	Infix	Suffix	LTPO	SoftCoT	TTSV
Tokens (mean)	575.7	558.4	549.9	573.3	592.3	594.4	577.4
Entropy (first 5%)	0.1332	0.1366	0.1402	0.1941	0.1662	0.4218	0.1359
Entropy (overall)	0.0871	0.0881	0.0899	0.1049	0.0909	0.1059	0.0864
Top-1 Prob (first 5%)	0.9535	0.9523	0.9525	0.9373	0.9437	0.9156	0.9534
Top-1 Prob (overall)	0.9684	0.9681	0.9678	0.9647	0.9683	0.9651	0.9685
Varentropy (first 5%)	0.2181	0.2207	0.2463	0.4010	0.2953	2.2234	0.2174
Varentropy (overall)	0.1353	0.1375	0.1422	0.2038	0.1452	0.2404	0.1361

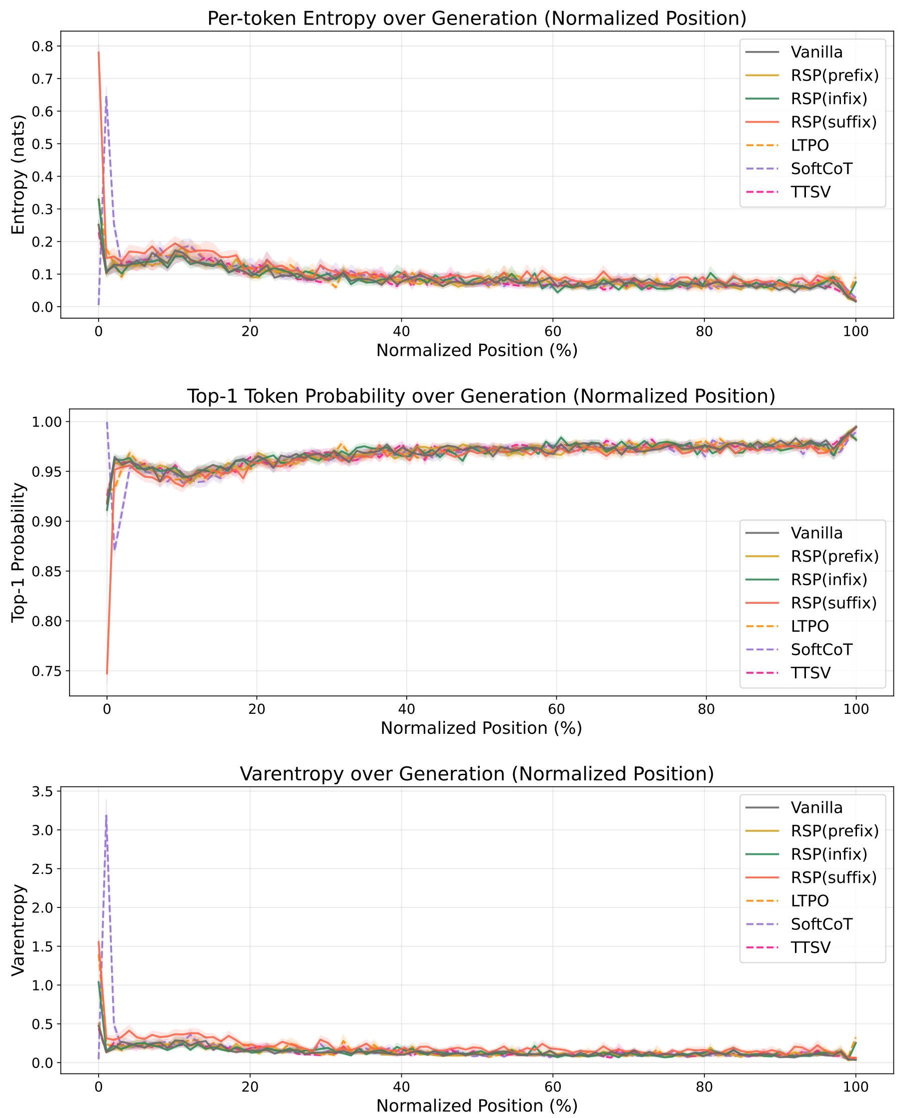


Figure 6: Mean entropy (top), top-1 probability (middle), and varentropy (bottom) over the full normalized generation trajectory (0–100%). Averaged over MATH-500, shading indicates ± 1 SEM. Solid lines: RSP variants and the baseline. Dashed lines: existing soft prompt methods. All methods converge after the initial stage.

B Token-Level Distribution Metrics

This appendix provides formal definitions for the metrics used in Section 4.4. Numerical results are reported in Table 3 of the main text. We extract first-token logits via a single forward pass (no autoregressive generation) and store the top-100 token ids and logit values per problem. Temperature conditions reuse the baseline logits scaled by $1/\tau$, requiring no separate inference. RSP draws 16 independent seeds per problem (MATH-500), and reported metrics are the mean across the 16 seeds. Pass@1 accuracy is computed over 16 samples per problem, with the baseline at τ=1.0 reaching $73.92 \pm 0.78\%$. Let P_{base} denote the baseline next-token distribution at $\tau = 1.0$ and P_{target} the candidate distribution being compared (e.g., baselines at τ=$2.0, 3.0$ or RSP at $\tau = 1.0$). All metrics are computed analytically from saved logits at the first generation step.

Spearman rank correlation. Spearman's ρ is the Pearson correlation coefficient between the token ranks of two distributions:

$$\rho \;=\; \mathrm{Pearson}(\mathrm{rank}(P_{\text{target}}),\; \mathrm{rank}(P_{\text{base}}))\,. \tag{4}$$

Because temperature scaling $P^{(\tau)} \propto P^{1/\tau}$ is a strictly monotone transform, the rank order of tokens is preserved exactly, so $\rho = 1$ for any $\tau > 0$ as a mathematical identity. Any value $\rho < 1$ therefore indicates rank reorganization beyond what temperature scaling can produce.

Mass outside top-K. Let $\mathrm{TopK}(P_{\text{base}})$ be the set of K tokens with the highest probability under P_{base}. The probability mass that the candidate distribution places *outside* this set is

$$\mathrm{MassOutside}_K \;=\; 1 - \sum_{t \in \mathrm{TopK}(P_{\text{base}})} P_{\text{target}}(t). \tag{5}$$

This directly measures support expansion: how much probability the candidate assigns to tokens that are not in the baseline's preferred set. Reported values use $K = 10$.

Jensen–Shannon divergence. The symmetrized, bounded version of Kullback–Leibler divergence:

$$\mathrm{JS}(P, Q) \;=\; \tfrac{1}{2}\,\mathrm{KL}(P \,\|\, M) + \tfrac{1}{2}\,\mathrm{KL}(Q \,\|\, M), \qquad M = \tfrac{1}{2}(P + Q). \tag{6}$$

We use base-2 logarithms so that JS $\in [0, 1]$. Tokens absent from the stored top-100 are assigned a sentinel logit (-10^9) before softmax, which is negligible for our setting.

C Prompt-Law Properties Used by the Theory

This appendix isolates the only properties of the RSP law used in the main text: centeredness, direction-agnostic covariance under a variance budget, and positive probability on every nonempty open set. We avoid stronger semantic claims because Section 3.4 needs only these geometric and measure-theoretic facts.

C.1 Centered Perturbations Do Not Impose Systematic First-Order Drift

Let Δ_{ij} be the baseline logit gap between tokens i and j, and write the first-order surrogate as

$$\Delta_{ij}(\bar{h}) = \Delta_{ij} + b_{ij}^{\top}\bar{h}.$$

Proposition 3 (No systematic first-order drift)**.** *If* $\mathbb{E}[\bar{h}] = 0$*, then*

$$\mathbb{E}\big[\Delta_{ij}(\bar{h})\big] = \Delta_{ij}$$

for every token pair (i, j)*. If a deterministic prompt* h_0 *is reused across runs, then*

$$\Delta_{ij}(h_0) - \Delta_{ij} = b_{ij}^{\top}h_0$$

is a fixed directional bias.

Proof. The centered case is immediate from linearity of expectation:

$$\mathbb{E}\big[\Delta_{ij}(\bar{h})\big] = \Delta_{ij} + b_{ij}^{\top}\mathbb{E}[\bar{h}] = \Delta_{ij}.$$

The deterministic case follows by substitution. □

This proposition rules out a run-averaged first-order bias. It does not say that individual prompt draws leave the logits unchanged.

C.2 Proof of Theorem 2

For a unit vector b, the first-order perturbation energy available along b is $\mathrm{Var}_{h\sim D}(b^{\top}h) = b^{\top}\Sigma_D b$. The worst-case directional energy is therefore the minimum Rayleigh quotient of Σ_D.

Proof. For any symmetric covariance matrix Σ_D,

$$\min_{\|b\|=1} b^{\top}\Sigma_D b = \lambda_{\min}(\Sigma_D).$$

Let the eigenvalues of Σ_D be $\lambda_1, \ldots, \lambda_d$. Then

$$\lambda_{\min}(\Sigma_D) \leq \frac{1}{d}\sum_{m=1}^{d}\lambda_m = \frac{\mathrm{tr}(\Sigma_D)}{d} \leq \frac{\rho^2}{d}.$$

Equality holds if and only if all eigenvalues are equal, i.e., $\Sigma_D = (\rho^2/d)\mathbf{I}_d$. □

The proof uses only covariance. We choose a Gaussian law for RSP because it adds the open-set reachability property proved next.

C.3 Open-Set Reachability of Isotropic Gaussian Prompts

Proposition 4 (Open-set reachability)**.** *Let* $\bar{h} \sim \mathcal{N}(0, \sigma^2\mathbf{I}_d)$ *with* $\sigma > 0$*. Then every nonempty open set* $U \subseteq \mathbb{R}^d$ *satisfies*

$$\Pr(\bar{h} \in U) > 0.$$

Proof. The Gaussian density

$$(2\pi\sigma^2)^{-d/2}\exp\left(-\frac{\|\bar{h}\|^2}{2\sigma^2}\right)$$

is strictly positive at every point of $\mathbb{R}^d$. Every nonempty open set contains a ball of positive Lebesgue measure, and integrating a strictly positive density over that ball gives positive probability. □

This is the only support property used in Proposition 1.

D Proofs for the Transformer-Level Mechanism

This appendix follows the same order as the main theory section: exploration, annealing, and performance gain. The prompt-law facts used before these proofs are collected in Appendix C.

D.1 Exploration: attention decomposition and perturbation size

Proposition 5 (Attention decomposition)**.** *For one attention head at query position i in layer ℓ, let $\alpha_{ij}^{(\ell)}$ be the attention weights over $n \geq 1$ unmasked real tokens and $L \geq 1$ random tokens, with finite attention logits. Define*

$$w_{r,i}^{(\ell)} := \sum_{j=1}^{L} \alpha_{i,n+j}^{(\ell)}$$

and, for $j \in [n]$,

$$\tilde{\alpha}_{ij}^{(\ell)} := \frac{\alpha_{ij}^{(\ell)}}{1 - w_{r,i}^{(\ell)}}.$$

Then

$$\mathbf{o}_i^{(\ell)} = (1 - w_{r,i}^{(\ell)})\, \tilde{\mathbf{o}}_i^{(\ell)} + \boldsymbol{\eta}_i^{(\ell)},$$

where

$$\tilde{\mathbf{o}}_i^{(\ell)} := \sum_{j=1}^{n} \tilde{\alpha}_{ij}^{(\ell)} \mathbf{v}_j^{(\ell)} \qquad \textit{and} \qquad \boldsymbol{\eta}_i^{(\ell)} := \sum_{j=1}^{L} \alpha_{i,n+j}^{(\ell)} \mathbf{v}_{r_j}^{(\ell)}.$$

Proof. The head output is

$$\mathbf{o}_i^{(\ell)} = \sum_{j=1}^{n} \alpha_{ij}^{(\ell)} \mathbf{v}_j^{(\ell)} + \sum_{j=1}^{L} \alpha_{i,n+j}^{(\ell)} \mathbf{v}_{r_j}^{(\ell)}.$$

By the softmax positivity and the existence of at least one unmasked real token, $\sum_{j=1}^{n} \alpha_{ij}^{(\ell)} = 1 - w_{r,i}^{(\ell)} > 0$, so

$$\sum_{j=1}^{n} \alpha_{ij}^{(\ell)} \mathbf{v}_j^{(\ell)} = (1 - w_{r,i}^{(\ell)}) \sum_{j=1}^{n} \frac{\alpha_{ij}^{(\ell)}}{1 - w_{r,i}^{(\ell)}} \mathbf{v}_j^{(\ell)} = (1 - w_{r,i}^{(\ell)})\, \tilde{\mathbf{o}}_i^{(\ell)}.$$

Substituting this identity gives the decomposition. □

Corollary 1 (Magnitude scales with random-token attention)**.** *For every realization of the random values,*

$$\|\boldsymbol{\eta}_i^{(\ell)}\| \leq w_{r,i}^{(\ell)} \max_{j \in [L]} \|\mathbf{v}_{r_j}^{(\ell)}\|.$$

Proof. By the triangle inequality,

$$\|\boldsymbol{\eta}_i^{(\ell)}\| = \left\| \sum_{j=1}^{L} \alpha_{i,n+j}^{(\ell)} \mathbf{v}_{r_j}^{(\ell)} \right\| \leq \sum_{j=1}^{L} \alpha_{i,n+j}^{(\ell)} \|\mathbf{v}_{r_j}^{(\ell)}\| \leq w_{r,i}^{(\ell)} \max_{j \in [L]} \|\mathbf{v}_{r_j}^{(\ell)}\|.$$

□

Corollary 1 is the only quantitative fact needed in the main text. No independence assumption between attention weights and random keys is required. Once $w_{r,i}^{(\ell)}$ is small, the random contribution must be small as well.

D.2 Corollaries of Theorem 1 on cache growth

The fixed-gap decay bound yields the corollaries below.

Corollary 2 (Sequence-wise annealing of the fixed-gap envelope)**.** *For fixed L and $\Delta_i^{(\ell)}$, the upper bound*

$$f(n) = \frac{L}{L + n\exp(\Delta_i^{(\ell)}/\sqrt{d})}$$

is strictly decreasing in n and tends to 0 as $n \to \infty$. Cache growth alone therefore narrows the largest RSP attention mass compatible with that gap.

During autoregressive decoding $\Delta_i^{(\ell)}$ also moves because queries, keys, and hidden states co-evolve. The accurate reading is not "the realized mass decreases at every step" but "the envelope compatible with a fixed gap shrinks as the cache grows."

Proof. Differentiate $f(n)$ with respect to n:

$$f'(n) = -\frac{L\exp(\Delta_i^{(\ell)}/\sqrt{d})}{\left(L + n\exp(\Delta_i^{(\ell)}/\sqrt{d})\right)^2} < 0.$$

Thus $f(n)$ is strictly decreasing, and its denominator diverges linearly in n while its numerator is fixed, so $f(n) \to 0$. □

Corollary 3 (Vanishing random contribution under annealing)**.** *If $n \to \infty$ while L and $\Delta_i^{(\ell)}$ remain fixed, $w_{r,i}^{(\ell)} \to 0$, and for every realization of the random values*

$$\|\boldsymbol{\eta}_i^{(\ell)}\| \le w_{r,i}^{(\ell)} \max_{j\in[L]} \|\mathbf{v}_{r_j}^{(\ell)}\| \to 0.$$

Proof. $w_{r,i}^{(\ell)} \to 0$ by Corollary 2. Apply Corollary 1 to bound $\|\boldsymbol{\eta}_i^{(\ell)}\|$. The maximum norm is finite for any realization of finitely many sampled values. □

D.3 Performance gain: from local admission to Pass@N

The main text ğ3.4 uses a first-order surrogate to state two claims: a baseline-excluded token can enter the local top-K set, and independent reruns can turn such local openings into Pass@N gains. The proofs below use only one support condition on the centered prompt law D:

$$\text{(S1)} \qquad D(U) > 0 \quad \text{for every nonempty open set } U \subseteq \mathbb{R}^d.$$

Proposition 4 shows that the isotropic Gaussian law of ğ3.4 satisfies (S1).

D.3.1 Autoregressive Formulation

In autoregressive decoding, the joint distribution over a sequence $y_{1:T}$ factorizes as

$$\Pr(y_{1:T} \mid x) = \prod_{t=1}^{T} p(y_t \mid x, y_{<t}),$$

where each step is governed by a prefix-conditioned distribution. Any perturbation introduced by RSP affects generation through a sequence of local changes to $p(y_t \mid x, y_{<t})$, rather than a single global distribution.

D.3.2 Local Linearization of Logits under RSP

Throughout this subsection, $\bar{h} \in \mathbb{R}^d$ denotes *one* centered prompt vector from the RSP definition (Eq. (1) in ğ3.1), treated as a per-token surrogate. The actual implementation uses a length-L prompt $\mathbf{H} = (h_1, \ldots, h_L)$ with L independent draws; the arguments below isolate how one local perturbation can shift the logits. We model the perturbed logits as a function $z_i(\bar{h})$, assuming local smoothness in a neighborhood of $\bar{h} = 0$:

$$z_i(\bar{h}) = z_i(0) + \nabla_h z_i(0)^\top \bar{h} + r_i(\bar{h}),$$

where $r_i(\bar{h}) = O(\|\bar{h}\|^2)$. Defining $a_i := \nabla_h z_i(0)$, we obtain the local affine surrogate

$$z_i^{\mathrm{lin}}(\bar{h}) := z_i + a_i^\top \bar{h}.$$

This approximation is used as a first-order surrogate for branch opening, not as an exact global model of the transformer logits.

D.3.3 Proof of Proposition 1

Let

$$\mathcal{G}_{i,K} \; := \; \big\{\bar{h} \in \mathbb{R}^d : \#\{j \neq i : z_j^{\mathrm{lin}}(\bar{h}) > z_i^{\mathrm{lin}}(\bar{h})\} \leq K - 1\big\}.$$

Proof. Assume $\mathcal{G}_{i,K}$ contains a nonempty open set U. By (S1), $D(U) > 0$. Since $U \subseteq \mathcal{G}_{i,K}$,

$$\Pr_{\bar{h}}(\bar{h} \in \mathcal{G}_{i,K}) \; \geq \; D(U) \; > \; 0.$$

This is exactly the event that token i enters the surrogate top-K set.

A useful sufficient condition is the existence of some $\bar{h}_0$ at which token i strictly outranks all but at most $K - 1$ other tokens in the affine surrogate. By continuity, a neighborhood of $\bar{h}_0$ is then contained in $\mathcal{G}_{i,K}$. □

D.3.4 Proof of Proposition 2

This proposition does not derive $p_{\min}(x) > 0$ from the mechanism. It only converts a task-side lower bound and independence into Pass@N growth.

Proof. Let A_n be the event that the n-th independent RSP run reaches a correct trajectory on problem x, and write $p_n(x) := \Pr(A_n) \geq p_{\min}(x)$ as in (M1). Under (M2),

$$\Pr\Big(\bigcap_{n=1}^{N} A_n^c\Big) = \prod_{n=1}^{N}(1 - p_n(x)) \leq \prod_{n=1}^{N}(1 - p_{\min}(x)) = (1 - p_{\min}(x))^N.$$

Taking complements gives

$$\Pr\big(\text{Pass@}N \text{ on } x\big) = \Pr\Big(\bigcup_{n=1}^{N} A_n\Big) \geq 1 - (1 - p_{\min}(x))^N.$$

The bound $1 - x \leq e^{-x}$ for $x \in [0, 1]$ then yields

$$1 - (1 - p_{\min}(x))^N \; \geq \; 1 - e^{-N p_{\min}(x)}.$$

□

E Attention Mass Analysis

This appendix formalizes the per-token attention mass reported in Figure 2 and the exclusion criteria applied to the reasoning span and the layer axis. For each of the 500 MATH-500 problems, a fresh RSP $\mathbf{H} \in \mathbb{R}^{L \times d}$ with $L = 10$ is sampled independently, so the reported heatmaps average over both problem variability and RSP-vector variability.

E.1 Per-Token Attention Mass

For each problem, we measure attention on the concatenated sequence [prompt; $\mathbf{H}$; generation], where the generation is the trajectory produced by the same model under matching RSP injection. Let $\alpha_{i,j}^{(\ell,h)}$ denote the post-softmax attention weight at layer ℓ, head h, from query position i to key position j, satisfying $\sum_j \alpha_{i,j}^{(\ell,h)} = 1$ under the causal mask. Let Q and R denote the index sets of question and RSP tokens with sizes $|Q|$ and $|R| = L$, and let H be the number of heads. The per-token attention mass that query i directs to each group at layer ℓ is

$$\mathrm{PTM}_Q[\ell, i] = \frac{1}{|Q|H} \sum_{h=1}^{H} \sum_{j \in Q} \alpha_{i,j}^{(\ell,h)}, \qquad \mathrm{PTM}_R[\ell, i] = \frac{1}{|R|H} \sum_{h=1}^{H} \sum_{j \in R} \alpha_{i,j}^{(\ell,h)}. \tag{7}$$

Normalization by $|Q|$ and $|R|$ controls for group size, so both quantities express how much attention a single question (resp. RSP) token receives on average under the same softmax denominator. Question tokens thereby serve as a size-matched baseline that absorbs sequence-length effects common to both groups.

E.2 Heatmap Aggregation

We partition the layer indices and the reasoning position indices each into five equal-size bins $\{B_L^{(b)}\}_{b=1}^5$ and $\{B_P^{(b)}\}_{b=1}^5$. For sample s and target group $\bullet \in \{Q, R\}$, the value in cell (b_L, b_P) is

$$M_\bullet^{(s)}[b_L, b_P] = \frac{1}{|B_L^{(b_L)}| \cdot |B_P^{(b_P)}|} \sum_{\ell \in B_L^{(b_L)}} \sum_{i \in B_P^{(b_P)}} \mathrm{PTM}_\bullet[\ell, i]. \tag{8}$$

Figure 2 reports the sample average of Eq. (8) over the 500 valid samples.

E.3 Excluding the `\boxed{...}` Span

The reasoning position range terminates strictly before the final `\boxed{...}` span. Chain-of-thought outputs decompose into a *text* (content) component and a *pattern* (template) component that contribute independently to downstream performance [Madaan and Yazdanbakhsh, 2022]. The `\boxed{...}` wrapper is a canonical pattern: a short, stereotyped span whose role is to complete the answer format rather than to extend reasoning. Including it would therefore inject a template-driven attention signature unrelated to reasoning dynamics.

E.4 Excluding the Final Two Layers

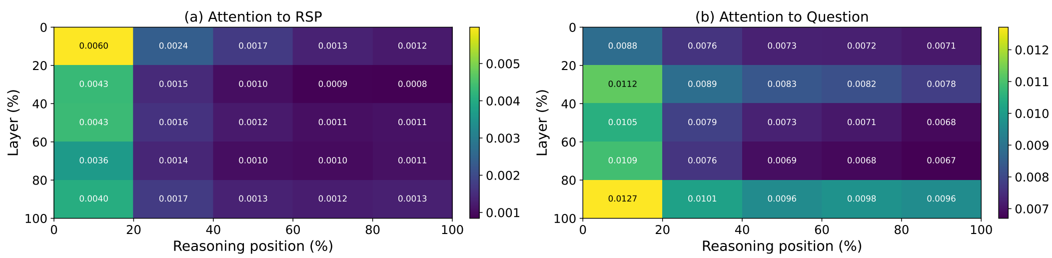
The layer axis further excludes the last two transformer layers. This choice is motivated by a standard late-layer effect rather than by any model-specific tuning decision.

Output-projection specialization. Late-layer hidden states lie in an approximately linear relationship with vocabulary logits and therefore function as a progressive linear readout rather than as representation-transforming blocks [Belrose et al., 2023]. Consistently, late-layer feed-forward modules have been characterized as output-distribution shaping mechanisms [Geva et al., 2021]. Attention in these layers therefore reflects output formation rather than reasoning computation.

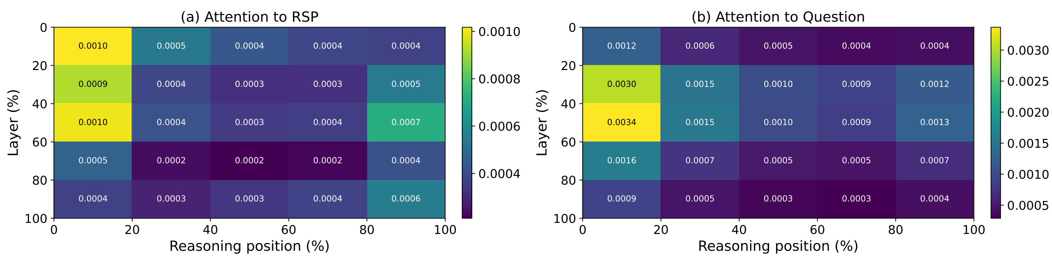
Excluding the final two layers keeps the heatmap focused on reasoning-phase dynamics instead of readout-phase dynamics. This exclusion is not load-bearing for the main claim: the sequence-direction attenuation emphasized in the main text is also visible without it, and Theorem 1 concerns sequence-direction attenuation rather than layer-wise monotonicity.

E.5 Additional Suffix Heatmaps

Figure 7 reports the same per-token RSP attention analysis under suffix injection for Qwen2.5-Math-1.5B-Instruct and LLaMA-3.1-8B-Instruct. For the late-layer reasons discussed above, the pattern does not generalize uniformly across every cell; nevertheless, the overall trend is consistent across both models: question-token attention varies broadly across layers, whereas RSP attention shows the layer-wise and sequence-wise decay predicted by Theorem 1.



(a) Qwen2.5-Math-1.5B-Instruct, suffix



(b) LLaMA-3.1-8B-Instruct, suffix

Figure 7: Per-token RSP attention mass under suffix injection for the remaining two models (500 MATH-500 samples each). Axes and preprocessing match Figure 2: x-axis is reasoning position binned into five quantiles, y-axis is layer depth binned into five quantiles (top = shallow), color encodes the 500-sample mean per-token attention assigned to RSP tokens, and tokens inside `\boxed{}` spans and the last two layers are excluded.

F DAPO Implementation

This appendix complements Section 5 with the DAPO loss modifications relative to GRPO, the DAPO + RSP implementation, and full per-step results.

F.1 From GRPO to DAPO

The vanilla GRPO objective [Shao et al., 2024] for a group of G rollouts $\{o_i\}_{i=1}^G$ with rewards $\{R_i\}_{i=1}^G$ is

$$\mathcal{J}_{\text{GRPO}}(\theta) = \mathbb{E}\left[\frac{1}{G}\sum_{i=1}^{G}\frac{1}{|o_i|}\sum_{t=1}^{|o_i|}\Big(\min\big(r_{i,t}\hat{A}_{i,t},\, \text{clip}(r_{i,t}, 1-\varepsilon, 1+\varepsilon)\hat{A}_{i,t}\big) - \beta\, D_{\text{KL}}(\pi_\theta \| \pi_{\text{ref}})\Big)\right], \tag{9}$$

with importance ratio $r_{i,t} = \pi_\theta(o_{i,t} \mid q, o_{i,<t})/\pi_{\theta_{\text{old}}}(o_{i,t} \mid q, o_{i,<t})$, group-relative advantage $\hat{A}_{i,t} = (R_i - \text{mean}(\{R_i\}))/\text{std}(\{R_i\})$, clipping range ε, and KL penalty $\beta\, D_{\text{KL}}$ against the reference π_{ref}.

We build on the SimpleRL-Zoo [Zeng et al., 2025] training pipeline and implement DAPO on top of VeRL [Sheng et al., 2024]. DAPO modifies Eq. (9) in four ways: (i) token-level loss aggregation $1/\sum_i |o_i|$ instead of $1/|o_i|$, (ii) asymmetric clipping with $(\varepsilon_{\text{low}}, \varepsilon_{\text{high}}) = (0.2, 0.28)$, (iii) dual clipping safeguard $\max(L_{\text{clipped}}, c\hat{A})$ with $c = 10$, and (iv) KL terms removed. The resulting DAPO objective is

$$\mathcal{J}_{\text{DAPO}}(\theta) \;=\; \mathbb{E}\left[\frac{1}{\sum_i |o_i|}\sum_{i,t}\min\Big(r_{i,t}\hat{A}_{i,t},\, \text{clip}\big(r_{i,t},\, 1-\varepsilon_{\text{low}},\, 1+\varepsilon_{\text{high}}\big)\hat{A}_{i,t}\Big)\right]. \tag{10}$$

F.2 DAPO + RSP Implementation

For DAPO + RSP, we additionally inject a group-specific RSP $\mathbf{H}_g \in \mathbb{R}^{L\times d}$ ($L = 20$) into each rollout. $\mathbf{H}_g$ is generated deterministically from a per-step group seed and injected via a forward hook; it is not a trainable parameter, so the learning signal is the policy adapting to arbitrary $\mathbf{H}_g$ rather than a learned prompt. Including $\mathbf{H}_g$ in the log-probabilities at both rollout and update time keeps the update on-policy with respect to the RSP-conditioned distribution (avoiding off-policy mismatch). With $G = 8$ and $n = 8$, each group contains a single response, so each rollout effectively sees a different $\mathbf{H}_g$.

F.3 Data, Batch, and Hyperparameters

Table 10: DAPO / DAPO + RSP training setup on Qwen2.5-Math-7B.

Field	**Value**
Train data	MATH Level 3–5 subset (`simplelr_math_35`, ~8,500 prompts)
Prompt template	`qwen-boxed`
Train batch size	1024
PPO mini-batch size	256 (4 PPO updates per rollout)
Rollouts per prompt G	8
Max prompt / response length	2,028 / 2,048
Rollout sampling	temperature 1.0, top-p 1.0, top-k 50
$(\varepsilon_{\text{low}}, \varepsilon_{\text{high}})$	(0.2, 0.28)
Dual clip c	10.0
Loss aggregation	token-mean ($1/\sum_i |o_i|$)
KL terms	disabled
Entropy coefficient	0
Optimizer	AdamW, learning rate 5×10^{-7} (constant), no warmup
Total training	~10 epochs, ≥ 80 optimization steps
Save / eval frequency	every 10 steps
RSP (DAPO + RSP only)	suffix injection, $L = 20$, freshly drawn per rollout
Hardware	$4\times$ NVIDIA B200 GPUs
Wall-clock training time	$\sim$12 hours per run

F.4 Evaluation Protocol

Each saved checkpoint is converted to HuggingFace format and evaluated using the SimpleRL-Zoo `eval_math_nodes.sh` pipeline, which selects sampling parameters per benchmark to reduce variance on the smaller competition sets:

Table 11: Per-benchmark evaluation protocol used by SimpleRL-Zoo `sh/eval.sh`.

Benchmark	# Problems	Temperature	n_{sampling}	Metric
AIME 2024	30	1.0	32	Avg@32
AMC 2023	40	1.0	32	Avg@32
MATH-500	500	0.0	1	accuracy (mean@1)
GSM8K	1,319	0.0	1	accuracy
OlympiadBench	675	0.0	1	accuracy
Minerva Math	272	0.0	1	accuracy
GaoKao 2023-EN	385	0.0	1	accuracy

The maximum number of generated tokens is 16,000 across all benchmarks. The reported average is the unweighted mean over the five benchmarks (GSM8K, MATH-500, College Math, Minerva Math, AIME 2024).

F.5 Per-Step Average Accuracy

Table 12: Five-benchmark average accuracy (%) at every checkpoint, computed over GSM8K, MATH-500, College Math, Minerva Math, and AIME 2024. **Bold** marks each method's peak.

Step	DAPO	DAPO + RSP
0	30.06	30.06
10	46.40	46.22
20	49.02	49.54
30	49.58	50.72
40	50.62	52.06
50	50.88	52.40
60	51.84	52.96
70	52.52	53.86
80	**53.20**	54.04
90	52.52	**54.36**
100	50.54	53.64

G Broader Impacts

This work provides empirical and theoretical analysis of how random embedding injection affects the reasoning behavior of large language models. The contributions are primarily foundational, with potential downstream implications for reinforcement learning-based post-training of reasoning models such as GRPO.

Positive impacts. RSP-induced trajectory diversity could improve the sample efficiency of RL-based LLM post-training by providing richer learning signals within group-relative advantage estimation. This has the potential to reduce compute costs associated with alignment and reasoning-oriented fine-tuning, making such methods more accessible to research groups with limited resources.

Potential concerns. Methods that expand the effective output support of LLMs may increase the reachability of low-probability tokens, which includes both desirable alternative reasoning paths and potentially unsafe or off-distribution outputs. Practitioners applying RSP in production systems should combine it with existing safety filtering and alignment mechanisms rather than treating it as a standalone diversity source.

Limitations on impact scope. Since this work focuses on analysis rather than deploying a new model or dataset, the direct societal impact is limited. The broader implications described above are contingent on future work integrating RSP into applied training pipelines.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and §1 state four scoped claims, each tied to a section: (i) RSP's contribution at each attention layer is captured by a single scalar with a fixed-gap upper-envelope cache-growth bound (Theorem 1, §3.3); (ii) under a local affine surrogate, isotropic re-sampling assigns positive probability to a top-K entry region unattainable by rank-preserving temperature sampling (Theorem 1, §3.4); (iii) RSP raises early-token entropy and stabilizes later, with Pass@N gains under independent re-sampling (§4); and (iv) the diversity transfers to RL training when combined with DAPO (§5). We do not claim a theorem that directly explains the format-mismatch recovery (Table 1); the wrong-attractor hypothesis there is explicitly informal.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Limitations are noted in several places rather than in a dedicated section: (a) Proposition 2 requires the task-side assumption $p_{\min}(x) > 0$, flagged in §3.4 and §6; (b) the $+18$–29 pp recovery on the format-mismatch rows of Table 1 is empirically observed but not formally derived from any theorem in §3, with the wrong-attractor account stated as a hypothesis to be quantified in future work (§4.2); (c) Table 1 reports the best-position accuracy across $\{$Prefix, Infix, Suffix$\}$, and per-position variance is large with several cases below baseline (Appendix A), so position is treated as a hyperparameter rather than something we justify principally; (d) DAPO + RSP results (§5) use a single training seed under a reduced DAPO configuration that omits Dynamic Sampling and overlong reward shaping; (e) main accuracy comparisons are single-seed without bootstrap or significance tests (item 7). Future work in §6 addresses training-time comparisons, domain expansion, and principled position selection.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [Yes]

   Justification: Theorem 1 (cache-growth attenuation) carries an inline proof in §3.3. Proposition 2, Theorem 1, and Proposition 2 are stated in §3.4 with proofs in Appendix D. Each statement explicitly lists its assumptions: finite logits and $n, L \geq 1$ for Theorem 1; locally affine surrogate around $\bar{h} = 0$ for Theorem 1; and $p_{\min}(x) > 0$ together with mutual independence of the N runs for Proposition 2.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

   Answer: [Yes]

   Justification: §4.1 and Appendix A specify benchmarks (MATH-500, GSM8K, AIME 2024 in the main table; further sets used for the DAPO study), models (LLaMA-3.1-8B-Instruct and Qwen2.5-Math 1.5B/7B variants), the RSP injection grid (Prefix/Infix/Suffix; $L \in \{10, 15, 20\}$), greedy decoding, and the answer extraction pipeline. Appendix F provides the full DAPO + RSP training recipe (batch of $1{,}024$ prompts, $G = 8$ rollouts per prompt at $\tau = 1.0$, four PPO mini-batch updates of 256 prompts each, 100 training steps, suffix RSP $L = 20$ during rollouts only).

5. **Open access to data and code**

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [No]

Justification: All datasets and base models used in the paper are publicly available and cited (MATH-500, GSM8K, AIME 2024, AMC 2023, OlympiadBench, Minerva Math, GaoKao 2023-EN, College Math; Qwen2.5-Math 1.5B/7B and LLaMA-3.1-8B-Instruct). RSP itself is described to a level of detail sufficient for re-implementation (Eq. (1) and §4.1). However, our RSP injection harness and DAPO + RSP training pipeline are not yet released; we will release an anonymized implementation alongside the camera-ready submission.

6. **Experimental setting/details**

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: §4.1 and Appendix A list all benchmarks, models, RSP injection settings (positions, lengths, per-row L choice in Appendix Table 7), maximum generation length per benchmark, and decoding strategy. Appendix F provides the full DAPO + RSP training settings, including optimizer, batch composition, mini-batch updates, asymmetric clipping range $(\varepsilon_{\text{low}}, \varepsilon_{\text{high}}) = (0.2, 0.28)$, dual clipping safeguard, and RSP injection protocol during rollouts.

7. **Experiment statistical significance**

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [No]

Justification: Pass@N experiments use $N = 16$ samples per problem (§4.4), Table 3 reports mean$\pm$standard deviation over 16 RSP seeds, and AIME 2024 is scored by Avg@32 to reduce variance. However, Table 1 (Table 1) reports single-run accuracies without bootstrap confidence intervals or paired tests, and the DAPO + RSP training in §5 uses a single training seed. Multi-seed evaluation, particularly on small competition sets (AIME 2024, AMC 2023) where variance is highest, is left for future work.

8. **Experiments compute resources**

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: Inference experiments (Tables 1, 2; Figure 3) run on a single NVIDIA A6000 40GB GPU with greedy decoding (Appendix A). DAPO + RSP training (§5) runs on $4\times$ NVIDIA B200 GPUs for approximately 12 hours per run (Appendix F, hardware row of the configuration table).

9. **Code of ethics**

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Answer: [Yes]

Justification: The work uses publicly available math reasoning benchmarks and pretrained language models, involves no human subjects or personal data, and does not introduce a release that bypasses standard model-safety considerations. We have reviewed the NeurIPS Code of Ethics and identify no conflicts.

10. **Broader impacts**

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: Appendix G discusses both directions: positive impact via more sample-efficient RL post-training of reasoning LLMs, and a potential downside that broadening the effective output support also increases reachability of undesirable trajectories, which we argue should keep RSP coupled with existing safety filtering rather than replacing it.

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

    Answer: [N/A]

    Justification: The paper does not release new pretrained models, generative systems, or scraped datasets. RSP is a generation-time perturbation applied to existing pretrained models and introduces no separate artifact requiring safeguards.

12. **Licenses for existing assets**

    Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

    Answer: [Yes]

    Justification: All datasets (MATH-500 [Lightman et al., 2024], GSM8K [Cobbe et al., 2021], AIME 2024, AMC 2023, OlympiadBench [He et al., 2024], Minerva Math [Lewkowycz et al., 2022], GaoKao 2023-EN, College Math) and pretrained models (Qwen2.5-Math [Yang et al., 2024], LLaMA-3.1 [Grattafiori et al., 2024]) are publicly available and cited at first use. The training pipeline builds on SimpleRL-Zoo [Zeng et al., 2025] and VeRL [Sheng et al., 2024], also cited. Each asset is used under its original publicly stated terms.

13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: [N/A]

    Justification: The paper does not release new datasets or pretrained models. The implementation code planned for camera-ready release (item 5) will be accompanied by configuration files and a README; no derived dataset or model checkpoint is part of the submission.

14. **Crowdsourcing and research with human subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: [N/A]

    Justification: The paper involves no crowdsourcing and no research with human subjects. All evaluations use static, publicly available math reasoning benchmarks.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

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    Justification: As noted in item 14, the paper involves no human subjects, so IRB review is not applicable.

16. **Declaration of LLM usage**

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Answer: [N/A]

Justification: Pretrained LLMs (Qwen2.5-Math, LLaMA-3.1) appear in the paper as the experimental subject of analysis, not as a non-standard component of the methods themselves. The methods (RSP definition, attention decomposition, DAPO + RSP training) do not rely on auxiliary LLMs in any non-standard or originality-affecting role.